

The Fundamental Theorem of Calculus



FTC Part 2: evaluating definite integrals

- 1 Evaluate $\int_0^3 (x^2 - 2x) dx$ using the Fundamental Theorem of Calculus.
 - 2 Evaluate $\int_1^4 \sqrt{x} dx$.
 - 3 Evaluate $\int_0^{\pi/2} \cos x dx$.
-

FTC Part 1: accumulation functions

- 4 Let $g(x) = \int_0^x t^2 dt$. Find $g'(x)$ using FTC Part 1.
 - 5 Let $g(x) = \int_1^x \sin(t^2) dt$. Find $g'(2)$.
 - 6 Let $g(x) = \int_0^{x^2} e^{-t} dt$. Find $g'(x)$ using FTC Part 1 combined with the Chain Rule.
-

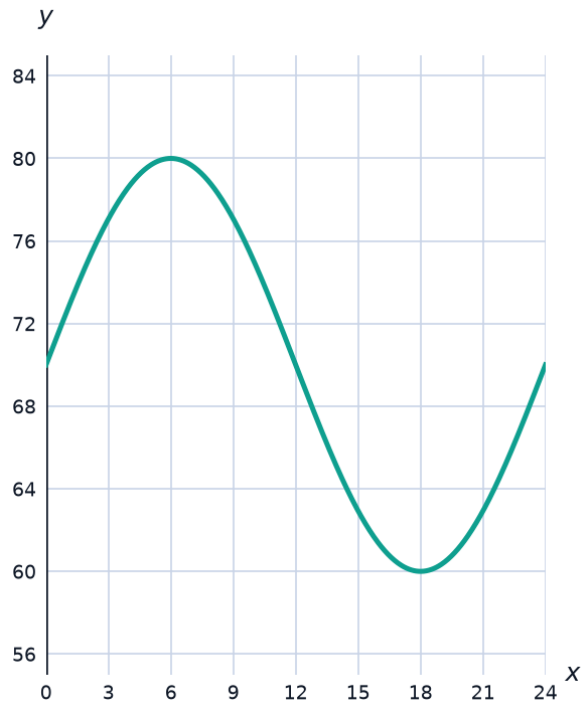
Interpreting accumulation functions

- 7 A particle's velocity is $v(t) = 3 - 2t$ (m/s). Let $s(t) = \int_0^t v(\tau) d\tau$ be its displacement from $t = 0$. Find $s(4)$ and interpret its meaning.
 - 8 Given the graph of f consists of straight-line segments with $f(0) = 0$, and $g(x) = \int_0^x f(t) dt$, explain what feature of g 's graph occurs where f crosses the x -axis, and where f has a local max.
-

Average value of a function

- 9 Find the average value of $f(x) = x^2$ on $[0, 3]$.

- 10 The temperature in a city over a 24-hour period is modeled by $T(t) = 70 + 10\sin\left(\frac{\pi t}{12}\right)$ for $0 \leq t \leq 24$. Find the average temperature over this period.



Definite integrals with substitution-friendly integrands

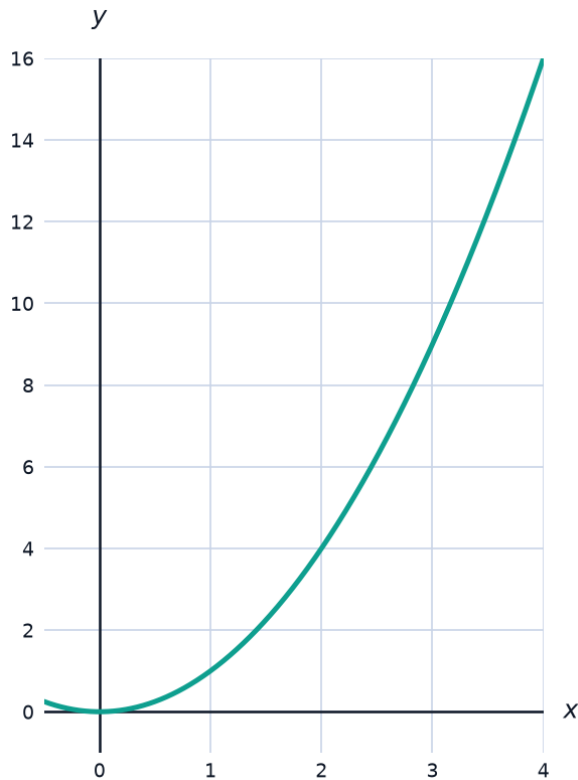
- 11 Evaluate $\int_0^1 2x(x^2 + 1)^3 dx$ using $u = x^2 + 1$.

The Mean Value Theorem for integrals

- 12 Find the value C guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$, i.e. the point where $f(C)$ equals the average value found earlier.

Visualizing FTC Part 1 as an accumulation function

- 13 The graph shows $f(t) = t^2$ (the integrand from $g(x) = \int_0^x t^2 dt$). Explain, using the graph, why g is always increasing, and why $g'(x) = f(x) = x^2$ matches the height of the curve at each x .



A capstone FTC application

- 14 Water flows into a tank at rate $r(t) = 8 - t$ liters/minute for $0 \leq t \leq 8$ (the rate decreases as the tank fills). Using FTC, find the total volume of water added during the first 6 minutes, and explain why the rate being zero at $t = 8$ is consistent with the tank reaching capacity.

FTC Part 1 with more complex bounds

- 15 Let $g(x) = \int_3^{2x} \ln(t+1) dt$. Find $g'(x)$ using FTC Part 1 and the Chain Rule.

A second average-value application

- 16 A company's hourly production rate over an 8-hour shift is $p(t) = 50 + 10t - t^2$ units/hour, for $0 \leq t \leq 8$. Find the average production rate over the shift.
-

Multiple choice — easy 15

17 Evaluate $\int_2^6 (6x^2 + 8x)dx$ using the Fundamental Theorem of Calculus.

- a. 544
- b. 546
- c. 576
- d. 542

18 Evaluate $\int_1^5 (6x^2 + 6x)dx$ using the Fundamental Theorem of Calculus.

- a. 320
- b. 318
- c. 322
- d. 325

19 Evaluate $\int_{-1}^0 (3x^2 + 2x)dx$ using the Fundamental Theorem of Calculus.

- a. -2
- b. 2
- c. 3
- d. 0

20 Evaluate $\int_2^4 (9x^2)dx$ using the Fundamental Theorem of Calculus.

- a. 170
- b. 168
- c. 166
- d. 192

- 21 Evaluate $\int_0^2 (-6x^2 - 2x)dx$ using the Fundamental Theorem of Calculus.
- a. -18
 - b. -22
 - c. -17
 - d. -20
- 22 Evaluate $\int_2^6 (6x^2 - 6x)dx$ using the Fundamental Theorem of Calculus.
- a. 318
 - b. 320
 - c. 322
 - d. 324
- 23 Evaluate $\int_{-2}^1 (3x^2 + 2x)dx$ using the Fundamental Theorem of Calculus.
- a. 4
 - b. 8
 - c. 2
 - d. 6
- 24 Evaluate $\int_2^7 (-3x^2 - 8x)dx$ using the Fundamental Theorem of Calculus.
- a. -513
 - b. -539
 - c. -517
 - d. -515

25 Evaluate $\int_2^7 (6x^2 + 2x)dx$ using the Fundamental Theorem of Calculus.

a. 715

b. 735

c. 717

d. 713

26 Evaluate $\int_0^3 (6x^2 - 2x)dx$ using the Fundamental Theorem of Calculus.

a. 45

b. 43

c. 48

d. 47

27 Evaluate $\int_{-2}^3 (6x^2)dx$ using the Fundamental Theorem of Calculus.

a. 70

b. 72

c. 54

d. 68

28 Evaluate $\int_{-2}^2 (-3x^2 - 2x)dx$ using the Fundamental Theorem of Calculus.

a. -12

b. -16

c. -14

d. -18

29 Evaluate $\int_0^5 (6x^2 + 4x)dx$ using the Fundamental Theorem of Calculus.

- a. 298
- b. 303
- c. 302
- d. 300

30 Evaluate $\int_1^4 (9x^2 + 6x)dx$ using the Fundamental Theorem of Calculus.

- a. 232
- b. 234
- c. 240
- d. 236

31 Evaluate $\int_2^5 (3x^2 - 6x)dx$ using the Fundamental Theorem of Calculus.

- a. 50
- b. 54
- c. 56
- d. 52

Multiple choice — medium 15

32 Let $g(x) = \int_{-1}^{-4x} t^2 dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

- a. 144
- b. -36
- c. -580
- d. -576

33 Let $g(x) = \int_{-3}^{-4x} t^2 dt$. Find $g'(2)$ using FTC Part 1 and the Chain Rule.

- a. -16
- b. -256
- c. 64
- d. -260

34 Let $g(x) = \int_{-1}^{-2x} t^2 dt$. Find $g'(1)$ using FTC Part 1 and the Chain Rule.

- a. -10
- b. -8
- c. -2
- d. 4

35 Let $g(x) = \int_0^{3x} t^2 dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

- a. 48
- b. 435
- c. 432
- d. 144

36 Let $g(x) = \int_1^{-3x} t^2 dt$. Find $g'(2)$ using FTC Part 1 and the Chain Rule.

- a. -12
- b. 36
- c. -111
- d. -108

37 Let $g(x) = \int_2^{-4x} t^2 dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.

a. -1028

b. 256

c. -1024

d. -64

38 Let $g(x) = \int_{-3}^{-3x} t^2 dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

a. -246

b. -243

c. -27

d. 81

39 Let $g(x) = \int_3^{2x} t^2 dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.

a. 74

b. 36

c. 18

d. 72

40 Let $g(x) = \int_1^{-3x} t^2 dt$. Find $g'(1)$ using FTC Part 1 and the Chain Rule.

a. -27

b. 9

c. -30

d. -3

- 41 Let $g(x) = \int_1^{-5x} t^2 dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.
- a. -2000
 - b. 400
 - c. -80
 - d. -2005
- 42 Let $g(x) = \int_2^{-3x} t^2 dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.
- a. 144
 - b. -432
 - c. -435
 - d. -48
- 43 Let $g(x) = \int_0^{-5x} t^2 dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.
- a. 225
 - b. -1125
 - c. -45
 - d. -1130
- 44 Let $g(x) = \int_{-3}^{4x} t^2 dt$. Find $g'(4)$ using FTC Part 1 and the Chain Rule.
- a. 64
 - b. 1028
 - c. 1024
 - d. 256

- 45 Let $g(x) = \int_{-2}^{2x} t^2 dt$. Find $g'(3)$ using FTC Part 1 and the Chain Rule.
- a. 74
 - b. 36
 - c. 72
 - d. 18
- 46 Let $g(x) = \int_3^{2x} t^2 dt$. Find $g'(1)$ using FTC Part 1 and the Chain Rule.
- a. 2
 - b. 10
 - c. 4
 - d. 8

Multiple choice — hard 15

- 47 Evaluate $\int_0^1 2x(x^2 + 2)^3 dx$ using $u = x^2 + 2$.
- a. $\frac{65}{4}$
 - b. $\frac{19}{4}$
 - c. $\frac{65}{3}$
 - d. $-\frac{65}{4}$
- 48 Evaluate $\int_0^1 2x(x^2 + 1)^2 dx$ using $u = x^2 + 1$.
- a. 1
 - b. $\frac{7}{3}$
 - c. $\frac{7}{2}$
 - d. $-\frac{7}{3}$

49 Evaluate $\int_0^1 2x(x^2 + 4)^3 dx$ using $u = x^2 + 4$.

a. $-\frac{369}{4}$

b. $\frac{369}{4}$

c. 123

d. $\frac{61}{4}$

50 Evaluate $\int_0^1 2x(x^2 + 4)^2 dx$ using $u = x^2 + 4$.

a. $\frac{61}{2}$

b. 3

c. $\frac{61}{3}$

d. $-\frac{61}{3}$

51 Evaluate $\int_0^1 2x(x^2 + 1)^3 dx$ using $u = x^2 + 1$.

a. $-\frac{15}{4}$

b. $\frac{7}{4}$

c. 5

d. $\frac{15}{4}$

52 Find the average value of $f(x) = -2x^2 - 4x + 5$ on $[0, 6]$.

a. -30

b. -91

c. -31

d. -32

53 Find the average value of $f(x) = x^2 + 4x + 3$ on $[0, 6]$.

a. 27

b. 63

c. 28

d. 26

54 Find the average value of $f(x) = x^2 - 4x$ on $[0, 2]$.

a. -4

b. $-\frac{5}{3}$

c. $-\frac{8}{3}$

d. $-\frac{11}{3}$

55 Find the average value of $f(x) = -3x^2 - 4x$ on $[0, 5]$.

a. -95

b. -36

c. -35

d. -34

56 Find the average value of $f(x) = 3x^2 + x - 3$ on $[0, 8]$.

a. 64

b. 66

c. 197

d. 65

- 57 Let $g(x) = \int_0^x f(t) dt$, where f is continuous and negative for $t < -2$ and positive for $t > -2$. What feature does g have at $x = -2$?
- a. An inflection point, since g is always increasing when its integrand is positive.
 - b. Neither -- accumulation functions never have local extrema.
 - c. A local minimum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.
 - d. A local maximum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.
- 58 Let $g(x) = \int_0^x f(t) dt$, where f is continuous and negative for $t < 5$ and positive for $t > 5$. What feature does g have at $x = 5$?
- a. An inflection point, since g is always increasing when its integrand is positive.
 - b. A local minimum, since f changes sign from negative to positive at $t = 5$, so $g' = f$ changes from $-$ to $+$.
 - c. A local maximum, since f changes sign from negative to positive at $t = 5$, so $g' = f$ changes from $-$ to $+$.
 - d. Neither -- accumulation functions never have local extrema.
- 59 Let $g(x) = \int_0^x f(t) dt$, where f is continuous and positive for $t < 1$ and negative for $t > 1$. What feature does g have at $x = 1$?
- a. Neither -- accumulation functions never have local extrema.
 - b. A local minimum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.
 - c. A local maximum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.
 - d. An inflection point, since g is always increasing when its integrand is positive.

- 60 Let $g(x) = \int_0^x f(t) dt$, where f is continuous and negative for $t < -5$ and positive for $t > -5$. What feature does g have at $x = -5$?
- a. A local minimum, since f changes sign from negative to positive at $t = -5$, so $g' = f$ changes from $-$ to $+$.
 - b. Neither -- accumulation functions never have local extrema.
 - c. A local maximum, since f changes sign from negative to positive at $t = -5$, so $g' = f$ changes from $-$ to $+$.
 - d. An inflection point, since g is always increasing when its integrand is positive.
- 61 Let $g(x) = \int_0^x f(t) dt$, where f is continuous and positive for $t < 5$ and negative for $t > 5$. What feature does g have at $x = 5$?
- a. A local maximum, since f changes sign from positive to negative at $t = 5$, so $g' = f$ changes from $+$ to $-$.
 - b. An inflection point, since g is always increasing when its integrand is positive.
 - c. A local minimum, since f changes sign from positive to negative at $t = 5$, so $g' = f$ changes from $+$ to $-$.
 - d. Neither -- accumulation functions never have local extrema.